

Supplement: Information-optimal illumination for dead-time-limited single-pixel imaging

[author block — user decision (mirrors main text)]

Numbering “Theorem 1–3” below refers to the main text; equations and propositions local to this supplement carry the S-prefix. All frozen constants trace to the R10/R11/R13/R14 rulings cited in the main-text source comments.

S1 The framework theorem: one master concave program

The framework theorem generalizes the main-text design program: with the random-hold kernel $J_H(\rho, \nu)$ of Theorem 2 inside the rank-one atoms $H_z = \nu J_H(\rho_z, \nu) q_z q_z^\top$, the design-measure problem remains concave, and every true interior optimum of the declared problem satisfies one generalized reduced-sensitivity/KKT system: the sensitivity $\psi(z) = M \operatorname{tr}[V^{-1} H_z] - \sum_m \beta_m c_m(z)$ obeys $\psi(z) \leq \eta$ with equality on support atoms, and at any support atom interior in a smooth scalar coordinate y , $\partial_y \psi = 0$ with $\partial_y^2 \psi \leq 0$. The load coordinate yields the information-water-filling equation — $M \nu J'_H(\rho) \ell$ equal to the marginal resource shadow price, with $\ell = q^\top V^{-1} q$ — while the support-scale and weight-power coordinates equate $2M \nu J_H(\rho) q_y^\top V^{-1} q$ to the corresponding marginal dose/peak/dispersion price: the apparent “conditioning penalty” is not an ad hoc term but the V^{-1} geometry inside the reduced sensitivity, which also explains why raising proxy contrast from $p = 1$ to $p = 2$ can lower the true objective. One caveat is kept deliberately: the master design problem places load, support scale, and intensity weighting in one constrained concave measure optimization, and its support atoms satisfy a common reduced-sensitivity condition; but the empirically observed occupancy ($k \approx 32$) and weighting ($p \approx 1$) optima *motivate* these coordinates and are not themselves claimed as continuous KKT theorems — they were selected from discrete grids and judged by image endpoints, not by the master log det objective.

S2 The alignment proposition

At a fixed design iterate with directional values $\ell_i = q_i^\top V^{-1} q_i$, allocate scalar loads under the budget $\sum_i \pi_i \rho_i = B$ to maximize $F(\rho) = \sum_i \pi_i \ell_i J(\rho_i)$; on an interval where J is increasing and strictly concave, the water-filling optimum satisfies $\ell_i J'(\rho_i^*) = \beta$ at every interior coordinate. A single global source induces $\tilde{\rho}_i = \kappa u_i$, with $u_i = a_i^\top x$ the bucket brightness.

Proposition S1 (Alignment and projected-residual bound). *Let $\pi_i > 0$, $\sum_i \pi_i = 1$, and consider*

$$\max_{\rho} F(\rho) = \sum_i \pi_i \ell_i J(\rho_i) \quad \text{subject to} \quad \sum_i \pi_i \rho_i = B, \quad \rho_i \in [\underline{\rho}_i, \bar{\rho}_i].$$

Assume $J \in C^2$ on the admitted interval and $-\ell_i J''(\rho) \geq m > 0$ for every i and every admitted ρ . Let the feasible fixed-flux allocation be $\tilde{\rho}_i = \kappa u_i$.

The fixed-flux allocation is optimal if and only if there exist a scalar β and normal-cone elements $s_i \in N_{[\underline{\rho}_i, \bar{\rho}_i]}(\tilde{\rho}_i)$ such that

$$\ell_i J'(\tilde{\rho}_i) - \beta - s_i = 0$$

for every i . For any β , choose s_i as a closest normal-cone element and define the projected KKT residual

$$r_i = \ell_i J'(\tilde{\rho}_i) - \beta - s_i.$$

$$0 \leq F(\rho^*) - F(\tilde{\rho}) \leq \frac{1}{2m} \sum_i \pi_i r_i^2. \quad (\text{S1})$$

If every $\tilde{\rho}_i$ is interior, $s_i = 0$ and $r_i = \ell_i J'(\kappa u_i) - \beta$, recovering the interior alignment condition.

Proof. The feasible set is convex and compact, and F is strictly concave under the curvature assumption, so the KKT conditions are necessary and sufficient and the optimum is unique. The equality multiplier is β ; the box constraints contribute the normal-cone elements s_i . This proves the first statement.

Let $\Delta_i = \rho_i^* - \tilde{\rho}_i$. Strong concavity of each $f_i(\rho) = \ell_i J(\rho)$ gives

$$f_i(\rho_i^*) - f_i(\tilde{\rho}_i) \leq f'_i(\tilde{\rho}_i) \Delta_i - \frac{m}{2} \Delta_i^2.$$

Write $f'_i(\tilde{\rho}_i) = \beta + s_i + r_i$ and sum with weights π_i . The budget equality gives $\sum_i \pi_i \Delta_i = 0$, while the normal-cone property gives $s_i \Delta_i \leq 0$. Completing the square term by term proves the stated bound; the lower bound follows from optimality of ρ^* . $\square$

A computable projected-KKT residual — not C_u alone — therefore certifies whether one global flux knob is close to the scalar water-filling optimum. In power-law form, if $J'(\rho) \asymp C\rho^{-s}$ and $\ell_i \asymp u_i^\alpha$, fixed flux aligns when $\alpha \approx s$; for the rising kernel $J = \rho/(1+\rho)$ the local slope $s(\rho) = 2\rho/(1+\rho)$ runs from 0 toward 2, so the often-invoked $\ell \propto u^2$ alignment is only a high-load approximation. Without the alignment condition there is no scene-independent constant-factor guarantee: two directions with $u_1/u_2 \rightarrow \infty$ but $\ell_1/\ell_2 \rightarrow 0$ drive the fixed-flux objective ratio to zero, so fixed flux can be arbitrarily bad. With random holds, J' vanishes at the jitter cap and is negative beyond it, so fixed flux can overdrive the brightest directions and explicit gain control becomes necessary. This explains both sides of the companion’s dev evidence [?]: RIDGE-SCAT32 with one global knob is near-optimal exactly when its realized residuals are small (self-water-filling), while the uniform servo — which loses by up to ~ 7 dB on sparse-support scenes — fails because forcing equal ρ_i discards a naturally favorable alignment between u_i and ℓ_i .

S3 Design machinery in full

S3.1 Task subspace and numerical ridge

The estimable subspace is frozen at $r = 200$: B is the top-200 eigenbasis of the fixed comparator’s information matrix under the same pre-scan, dwell, and resource convention, computed before the optimization and held fixed, with a fixed numerical ridge $\epsilon_0 = 10^{-9} \text{tr}(B^\top V_{\text{FIXED}} B)/r$ (R13 §6). A design-dependent ridge is not permitted: it would change the objective derivative and invalidate the sensitivity calculation.

S3.2 The full-stack near-optimality certificate

By concavity of log det and affineness of $\xi \mapsto V(\xi)$, a feasible design ξ^* is optimal iff there exist constraint multipliers such that the reduced sensitivity $s(a) = \text{tr}[V(\xi^*)^{-1} H(a)] - \sum_j \alpha_j e_j(a) - \sum_k \beta_k c_k(a)$ satisfies $s(a) \leq \eta^*$ over $\mathcal{A}$ with equality on the support (the constrained Kiefer–Wolfowitz / general equivalence theorem for nonlinear models [?, ?]). The confirmatory certificate evaluates this gap for the *deployed* balanced SCAT32 design against the continuous optimum of the *full deployed conditioning stack* over the expanded dictionary $D_{\text{cert}} = D_{\text{load}} \cup D_{\text{gain}}$ (§??). With $V(\xi) = V_{\text{pre}} + M_{\text{main}} \sum_a \xi_a H_a + \epsilon_0 I$ on the frozen $r = 200$ subspace and V_{fix} the load-matched deployed SCAT32 information matrix for the same pre-scan, dwell, budget, subspace, and numerical ridge, the certified class is

$$\mathcal{C}_{\text{stack}} = \left\{ \xi : \begin{array}{l} \xi \geq 0, \quad \mathbf{1}^\top \xi = 1, \quad c^\top \xi \leq b, \quad U^+ \xi \leq 0, \quad U^- \xi \leq 0, \\ \text{tr}\{V(\xi)^{-1}\} \leq 1.05 \text{tr}(V_{\text{fix}}^{-1}), \quad V(\xi) \succeq 0.5 V_{\text{fix}} \end{array} \right\}, \quad (\text{S2})$$

where $U^\pm$ encode the $\pm 5\%$ per-pixel dose band. Pointwise peak, ceiling, support, weight, and RQL-trust restrictions remain atom-admission guards; the A-risk and spectral conditions are *global design constraints* and are never approximated by dropping individual atoms. With atom sensitivity $d_a = M_{\text{main}} \text{tr}(V_d^{-1} H_a)$ at

the deployed matrix V_d and $\bar{d}_d = \text{tr}[V_d^{-1}(V_d - V_{\text{pre}} - \epsilon_0 I)]$, concavity gives $\log \det V(\xi) - \log \det V_d \leq d^\top \xi - \bar{d}_d$ for every feasible ξ , so the support-function bound

$$G_{\text{stack}} = \sup_{\xi \in \mathcal{C}_{\text{stack}}} \{d^\top \xi - \bar{d}_d\}, \quad G_{\text{stack}}/r \leq 10^{-2}, \quad (\text{S3})$$

upper-bounds the optimality gap and certifies local D-efficiency $\text{Eff}_D \geq \exp(-G_{\text{stack}}/r) \geq e^{-0.01} = 0.99005$. The spectral constraint enters exactly as the linear matrix inequality $V(\xi) - 0.5 V_{\text{fix}} \succeq 0$, and the A-risk cap through the Schur block $\begin{bmatrix} V(\xi) & I \\ I & W \end{bmatrix} \succeq 0$ with $\text{tr} W \leq 1.05 \text{tr}(V_{\text{fix}}^{-1})$; the dual of the resulting conic linear program carries one nonnegative budget multiplier, upper/lower pixel-dose multipliers, a positive-semidefinite matrix price for the spectral LMI, a positive-semidefinite block price and trace multiplier for the A-risk LMI, and a full-dictionary maximum reduced sensitivity. The frozen implementation is a certified cutting-plane outer approximation: spectral eigenvector cuts $z^\top \{V(\xi) - 0.5 V_{\text{fix}}\} z \geq 0$ for every discovered violating generalized eigenvector z ; A-risk tangent cuts $h(V_k) - \text{tr}\{V_k^{-2}(V - V_k)\} \leq 1.05 h(V_{\text{fix}})$ for the convex $h(V) = \text{tr}(V^{-1})$ with gradient $-V^{-2}$; a linear master with budget/dose prices and coverage-seeded column generation [?]; and an exact scan over all of D_{cert} each round. A finite cut set defines an *outer* relaxation of $\mathcal{C}_{\text{stack}}$, so a small master bound remains conservative; a large bound is never called a counterexample unless a feasible primal design is constructed. To that end every cell also runs a frozen *support-expanding* primal search over $\mathcal{C}_{\text{stack}}$ (positive initialization on all admitted columns or Frank–Wolfe atom injection); a feasible design with actual $[\log \det V(\xi) - \log \det V_d]/r > 10^{-2}$ classifies the cell **COUNTEREXAMPLE**, and a primal lower bound never certifies near-optimality — it can only refute it. Each of the 480 cells receives exactly one terminal status — **CERTIFIED**, **COUNTEREXAMPLE**, or **NUMERICAL_UNRESOLVED** — under the frozen compute protocol (60 s primal screen, 180 s primary cutting-plane solve, one deterministic rescaled retry of 180 s under frozen equilibration; **CERT_CELL_WALL_CAP** = 420 s; no third attempt, no threshold change, no imputation, no cell deletion), and records solver status (primary and retry), wall seconds, coefficient dynamic range, dictionary-scan and A-risk/spectral cut counts, the full-dictionary scan residual, primal and dual gaps per r , the minimum generalized eigenvalue, the A-risk ratio, and **MU_CAP_ACTIVE** (an active finite dual cap certifies only after a $\times 100$ cap-stability rerun changes the bound by at most $10^{-4}r$ with all residuals still passing). Before freeze the pipeline must pass a four-toy suite (budget+dose active; A-risk only; spectral only; all four active with a dose-feasible design rejected by conditioning) with independent primal solves, primal–dual agreement $\leq 10^{-8}$, minimum PSD residual $\geq -10^{-9}$, normalized complementarity $\leq 10^{-8}$, and full-dictionary scan residual $\leq 10^{-8}$. The certificate is comparator- and subspace-relative by construction: it preserves the predeclared conditioning of SCAT32 on the directions SCAT32 resolves, and it is not a global optimum over all images, subspaces, risk tradeoffs, or physical patterns. Continuous weights are rounded to an exact 972-row design by efficient rounding followed by deterministic exchange [?, ?, ?], admissible only if the exact D-efficiency ≥ 0.95 .

S3.3 Guards as constraints

The raw optimum is admissible only after all frozen guards pass; each is a constraint of the master program or a post-rounding check, not a tunable proxy. *Peak (two-part, R13 A2)*: a shape guard bounds unit-load support weights to $\frac{1}{4} \leq w_j \leq 4$ with support size $k \geq 16$, and a physical guard caps $\max_{i,j} a_{ij}^{\text{physical}} \leq 1536$ (the peak of a uniform $k = 16$ atom at the absolute load cap $\rho = 24$), so matched-intensity ridge atoms cannot silently demand more peak power. *Weight power (R10 G4)*: the dictionary contains powers $p \in \{0, 1\}$ only; $p = 2$ is excluded because dev evidence showed higher proxy contrast can worsen conditioning and endpoint quality. *Per-pixel dose (R10 G5)*: each pixel’s mean exposure stays within $\pm 5\%$ of the image mean, enforced in-optimizer and re-verified after rounding; a necessary row bound follows, $\rho_r \leq 1.05 M \bar{d} / \max_j g_{rj}$ (R13 A5). *Spectral floor (R10 G6)*: $V_{\text{OED}} \succeq 0.5 V_{\text{FIXED}^*}$ on B , with a convex-mixture fallback whose coefficient is fixed from information matrices before any image endpoint is seen. *Kernel and trust (R11 G7–G8, R13 A1)*: every (ν, ρ) node is certified for exact-PMF mass, score, and derivative agreement, mean-information efficiency $J_{\text{mean}}/J_{\text{exact}} \geq 0.90$, finite-window RQL log-load bias $|\log(\rho^\dagger/\rho)| \leq 0.05$, ceiling probability $p_{\text{ceil}}(\rho, \nu) \leq 0.05$ per atom and ≤ 0.01 design-weighted. *A-risk (R10 G7)*: $\text{tr}(V_{\text{exact}}^{-1}) \leq 1.05 \text{tr}(V_{\text{FIXED}^*}^{-1})$. Any violation is a declared failure state (**DICTIONARY_RANK_FAIL**, **CONTINUOUS_CERTIFICATE_FAIL**, **SPECTRAL_GUARD_FAIL**, **A_RISK_GUARD_FAIL**, **DOSE_GUARD_FAIL**, **ROUNDING_FAIL**), whose predeclared fallback is the fixed comparator; no threshold is retuned after freeze.

S3.4 The frozen dictionary

The exact searchable base dictionary D_{load} includes all cyclic translates of: scattered balanced supports at $k = 16$ and $k = 32$; the solid 4×4 patch; the committed LBLOB 6×6 16-pixel support; bars/rectangles 1×16 , 2×8 , 4×4 , 8×2 , 16×1 ; compact 32-pixel square/rectangle variants; and the frozen 16-pixel ring/annulus family — with $p = 0$ and guarded $p = 1$ weights for every applicable support. The implementation supports variable k ; all translation families use the circulant/FFT evaluator, and the dictionary manifest carries every family, translation, power, support size, and SHA256. These are the target-load-normalized atoms: each pattern’s amplitude is chosen to hit a predicted load.

For the near-optimality certificate only, an additional gain-coupled family D_{gain} is formed: for every frozen base geometry and weight pattern, physical rows $a_{g,\gamma} = \gamma a_g^{\text{base}}$ with $\gamma \in \{0.2, 0.5, 1, 2, 5\}$, the base a_g^{base} normalized by the scene-independent physical convention of the fixed patterns, and predicted load allowed to *emerge* as $\rho_{g,\gamma} = \bar{\rho} \gamma (a_g^{\text{base}})^\top \hat{x}$ — never rescaled to a target load, so fixed-flux self-water-filling and bounded per-pattern gain control lie inside the comparison class. The same support, weight, physical-peak, ceiling, and RQL-trust admission guards apply to both parts; $D_{\text{cert}} = D_{\text{load}} \cup D_{\text{gain}}$ is SHA256-frozen before any confirmatory scene is opened.

S3.5 Budget accounting

For every arm both the incident budget $B_{\text{inc}} = \sum_i \lambda_{s,i} T$ and the detected budgets $B_{\text{det}}^{\text{pred}} = \sum_i \mathbb{E}[N_i]$, $B_{\text{det}}^{\text{obs}} = \sum_i N_i$ are reported, together with total and maximum per-pixel dose. B_{det} may serve as an electronics-throughput guard if a hardware limit is frozen before confirmation, but it is never the sole fairness constraint: for a nonparalyzable detector $\mathbb{E}[N \mid \rho] \simeq \nu \rho / (1 + \rho)$, whose marginal cost tends to zero at high load, so a detected-count-only budget can permit arbitrarily large incident power while appearing nearly cost-free.

S4 Monte-Carlo verification detail

The refined sweep (`jitter_sfi.v2`: lognormal holds, 100,000 frames per cell, common random numbers, 25-point log grid of spacing 1.157, $\nu \in \{200, 2000\}$) confirmed the R14 law out-of-sample. At $\nu = 2000$ the measured peaks are $\rho_* = 22.297, 10.739, 5.173, 3.862, 2.153$, and 1.607 at $c_v = 0, 0.02, 0.05, 0.1, 0.2$, and 0.3, respectively; both predictions registered before the data existed hit — $c_v = 0.02$ predicted ≈ 10.4 , measured 10.739 (3%); $c_v = 0.2$ predicted ≈ 2.30 , measured 2.153 (6%) — and every in-sample peak lies within one grid notch of its exact-law value (commits `8a627bb`, predictions registered pre-data; `1d8d7aa`, verdict). The $\nu = 200$ column is consistent with the crossover correction: the $c_v = 0$ peak at 9.28 tracks the exact ridge 10.04, and $c_v = 0.3$ measures 1.86 against the predicted 1.77.

A continuous-peak zoom (quadratic interpolation, local run, 150,000 frames) and an independent replication (Colab, 400,000 frames) then measured the four jittered peaks twice each at $\nu = 2000$: local 9.645, 5.154, 3.752, 2.106 and replicated 9.610, 5.700, 3.399, 2.051 at $c_v = 0.02, 0.05, 0.1, 0.2$, respectively. The pooled eight-measurement log-log fit gives slope -0.658 vs. the predicted $-2/3$ (1.3%); the constant sits about 5% below $2^{-1/3}$, at the edge of the run-to-run noise, so the frozen statement is: exponent confirmed at 1.3%, constant consistent within 5–10%. The end points replicate a mild $\approx -8\%$ low bias across both runs, left as an open higher-order note (candidate: a lognormal third-moment correction to the leading finite-variance law). One intermediate hypothesis is on the record: after the first two zoom points, an effective jitter-cost multiplier $k = 4/3$ was preregistered (commit `5e51088`) and was refuted by the very next measured point, whose deviation flipped sign (commit `1a32758`) — peak-location noise on flat information tops, not a systematic constant. The fuller three-fold-family exponent battery frozen in the ruling (tests E1–E5, with matched-CV gamma and bounded two-point holds and cross-fitted conditional-score estimation) remains a preregistered extension beyond this verdict.

Table S1: Per-dwell ridge operating-point results (median over the 24 confirmatory scenes; source: imaging shards). ΔQ columns are median seed-mean radiometric-PSNR gains; $n_{>0}$ counts scenes with positive gain against SCAT32-060. Only the $\nu = 2000$, SCAT32-060 entry is the frozen primary gate (+1.87 dB, 19/24).

ν	achieved $\bar{\rho}$	ΔQ vs 060 (dB)	ΔQ vs SAFE (dB)	$n_{>0}$ vs 060
5	0.246	-0.08	+0.15	2/24
10	0.487	-0.03	+0.25	8/24
20	0.967	+0.10	+0.65	24/24
50	2.401	+0.34	+1.16	24/24
100	4.778	+0.76	+1.70	24/24
200	9.398	+1.14	+1.97	22/24
500	13.806	+1.31	+2.99	21/24
1000	17.541	+1.71	+3.65	20/24
2000	22.255	+1.87	+4.56	19/24

S5 Secondary-arm and certificate full results

S5.1 Ridge operating-point and time-to-quality

Table ?? gives the per-dwell fixed-dwell median radiometric gains of RIDGE-SCAT32 over the primary comparator SCAT32-060 and over the safe reference SCAT32-SAFE, pooled over the 24 confirmatory scenes (ΔQ is the seed-mean per-scene gain, median over scenes; only the $\nu = 2000$ row against SCAT32-060 carries the primary gate). At every dwell the ridge policy reaches its target mean load $\rho_R(\nu)$ exactly: the achieved mean load equals the ridge target (maximum 22.25 at $\nu = 2000$, below the absolute cap 24), so the global safety clip never binds and no `RIDGE_GUARD_CLIPPED` event is recorded; the predicted count-ceiling fraction $p_{\text{ceil}} < 10^{-6}$ and the realized dark-count/timing-error fractions are 0 at every dwell. The ridge advantage over SCAT32-060 grows with dwell (crossing positive near $\nu = 20$) and peaks near $\nu = 1000$ –2000; the advantage over the 0.05 safe reference is larger at fixed dwell. This fixed-dwell picture does Using the preregistered elapsed optical-time coordinate $T_{\text{opt}} = M_{\text{total}}\nu\tau$ and the frozen Q90/PAVA/censoring procedure, the corrected ridge time-to-quality endpoint passed: median $S_{\text{gate}} = 19.127043091646133$, family-stratified nested-bootstrap 2.5th-percentile bound 18.328492357080282, and 21/24 scenes above one.

S5.2 Descriptive full-stack analysis distribution

The preregistered 24-cell development feasibility screen selected the `FALLBACK_DESCRIPTIVE` branch before freeze. The post-freeze full-stack computation is therefore descriptive and has no PASS/FAIL endpoint. It returns over the 480 frozen cells (D_{cert} SHA 908cfccbd249de22). The terminal-status distribution is 0 `CERTIFIED`, 299 `COUNTEREXAMPLE`, 181 `NUMERICAL_UNRESOLVED`, stable across the four anchors (ordered `CERTIFIED/COUNTEREXAMPLE/UNRESOLVED`: $(\nu, b) = (200, 0.05)$: 0/77/43; $(200, 0.60)$: 0/67/53; $(2000, 0.05)$: 0/84/36; $(2000, 0.60)$: 0/71/49). Every cell is deployed-design feasible (`dep_feasible` true in all 480; realized per-pixel dose deviation 0.040, inside the $\pm 5\%$ band) and none has an active finite dual cap (`MU_CAP_ACTIVE` false in all 480). The 299 counterexamples are found by the frozen 60 s support-expanding primal screen (`solver_status_primary` = `SKIPPED_COUNTEREXAMPLE`, the dual solve short-circuited), each a full-stack-feasible design with actual per-dimension log-determinant improvement (`primal_gap_lower_per_r`) over deployed SCAT32 in the range 1.19–1.87 (median 1.43), two orders of magnitude above the 10^{-2} bar. The 181 unresolved cells return an infeasible cutting-plane master under the frozen numerical protocol (`LP_FAIL` on both the primary solve and the single deterministic rescaled retry; coefficient dynamic range 4000), so no finite dual upper bound G_{stack}/r is obtained for them; per the taxonomy these are *not certified* rather than refuted. Wall time is well inside budget (median 64.6 s, maximum 67.3 s against the 420 s `CERT_CELL_WALL_CAP`; no cell reaches the cap; no imputation, no cell deletion). Because no cell is certified, the branch is descriptive: no dual/complementarity near-optimality bound is asserted, and the counterexample cohort is reported as confirmatory descriptive evidence that scene-adapted geometry headroom survives the full deployed conditioning stack, with no categorical collapse or near-optimality claim.

S5.3 Context arms and design diagnostics

At the deployed operating point ($\bar{\rho} = 0.60$, $\nu = 2000$) the frozen context ladder places the deployed $k = 32$ scattered support above both descriptive references at matched load: mean radiometric PSNR 17.12 dB (SCAT32-060) $>$ 14.18 dB (SCAT16, scattered $k = 16$) $>$ 9.63 dB (LBLOB16, clustered), consistent with the frozen development-selection rule (main-text Section on the deployed design). The dose-relaxed and dose-constrained continuous OED solutions are retained as no-endpoint design diagnostics feeding the Act III figure; the development-cell guard-versus- α scan terminates at zero adaptive rows under the joint dose/A-risk/spectral guard stack (ADAPTIVE_COLLAPSE_UNDER_GUARDS), its only surviving descriptive artifact being the path-specific feasibility index PATH_FEASIBLE_ALPHA, taken relative to the frozen row ordering and never an estimator, fallback, or certificate. The dose-only class is retained only as a no-gate development analysis with source-of-record fields DOSE_ONLY_PRIMAL_GAP_PER_R, DOSE_ONLY_D_EFFICIENCY_LOWER, DOSE_ONLY_DUAL_UPPER_IF_AVAILABLE, and DOSE_ONLY_SOLVER_STATUS; its current values are lower bounds on available headroom, not estimates of the global dose-only optimum, and no confirmatory dose-only verdict exists.

S6 Proofs

Installed proofs. Theorem ?? follows from the complete-path martingale score and the law of total variance. Theorem ?? uses the delayed-renewal local normal approximation under the regularity stated below, and Theorem ?? follows by differentiating the limiting information rate. The alignment result uses a projected KKT residual at box boundaries. The crossover proposition is conditional on a uniform C^2 triangular-array expansion; its matched-asymptotic-prediction label is retained.

Proposition S2 (Uniform crossover constant from the triangular expansion). *Let $t = \nu^{-1/3}$, $x = 12\nu c_v^2$, and $\rho = t^{-1}y$. Assume that, uniformly on compact x - and y -sets containing $y_0(x) = (6/(1+x))^{1/3}$,*

$$J_{\nu, c_v}(t^{-1}y) = 1 - tF_1(y, x) + t^2F_2(y, x) + R_t(y, x),$$

where $F_1 = y^{-1} + (1+x)y^2/12$, $F_2 = y^{-2} + (x-2)y/12 + (1+x)^2y^4/144$, and R_t and its first two y -derivatives are $O(t^3)$. Then the principal maximizer is unique for sufficiently large ν and

$$\rho_* = (2c_v^2 + (6\nu)^{-1})^{-1/3} - \frac{12\nu c_v^2 + 4}{6(12\nu c_v^2 + 1)} + O(\nu^{-1/3}).$$

Proof. $F'_1 = -y^{-2} + (1+x)y/6$ has the unique minimizer y_0 and $F''_1(y_0, x) = (1+x)/2 > 0$. The uniform C^2 remainder and the implicit function theorem give $y_* = y_0 + ty_1 + O(t^2)$, with $y_1 = F'_2(y_0, x)/F''_1(y_0, x) = -(x+4)/(6(x+1))$. Substitution of $x = 12\nu c_v^2$ and $t = \nu^{-1/3}$ gives the displayed expansion. $\square$

Proof of Theorem 1 (missing-information identity). Let A_t be the predictable detector-active indicator and $L_T = \int_0^T A_t dt$. Conditional on the state path, the score for $\theta = \log \lambda$ is $U_T = N_T - \lambda L_T$. The compensated count is a martingale with quadratic variation λL_T , hence $I_{\text{path}} = \mathbb{E}N_T$. For $\mathcal{G}_T = \sigma(N_T)$, $U_N = \mathbb{E}(U_T \mid \mathcal{G}_T)$, and total variance gives

$$I_N(\theta; T) = \mathbb{E}N_T - \lambda^2 \mathbb{E}[\text{Var}(L_T \mid N_T)].$$

$\square$

Proof of Theorem 2 (long-window rate). Rescale by τ , so $\mathbb{E}B = 1$, $\text{Var } B = c_v^2$, and an ordinary cycle is $C = B + X$ with $X \sim \text{Exp}(\rho)$. Thus $\mu_C = 1 + \rho^{-1}$ and $\sigma_C^2 = c_v^2 + \rho^{-2}$. Delayed-renewal asymptotics give $m_\nu = \nu\rho/(1+\rho) + O(1)$, $v_\nu = \nu\rho(1+c_v^2\rho^2)/(1+\rho)^3 + O(1)$, and $\dot{m}_\nu = \nu\rho/(1+\rho)^2 + O(1)$. Under the regular nonlattice local-limit/LAN assumptions, $I_N = \dot{m}_\nu^2/v_\nu + O(1)$, so $I_N/\nu \rightarrow J_\infty = \rho/[(1+\rho)(1+c_v^2\rho^2)]$. Subtracting this from the complete-path rate $\rho/(1+\rho)$ gives the stated missing rate. $\square$

The local-limit step is invoked for a regular nonlattice delayed-renewal family with a finite $3 + \epsilon$ moment and parameter derivatives controlled on compact load intervals; lognormal, gamma/exponential, and bounded regular hold families satisfy the intended conditions. The theorem is not asserted for infinite-variance, correlated, observed, or load-dependent holds.

Proof of Theorem 3 (cubic optimum). For $J_\infty = \rho/[(1+\rho)(1+c_v^2\rho^2)]$,

$$\frac{d}{d\rho} \log J_\infty = \frac{1 - c_v^2 \rho^2 (1 + 2\rho)}{\rho(1+\rho)(1+c_v^2\rho^2)}.$$

The numerator has one zero, which is the unique global maximum because the rate vanishes at both endpoints. Setting $\varepsilon = c_v^{2/3}$ and solving $\varepsilon^3 \rho^2 (1 + 2\rho) = 1$ by powers gives $\rho = 2^{-1/3} c_v^{-2/3} - 1/6 + 2^{1/3} c_v^{2/3}/36 + O(c_v^{4/3})$. $\square$

Proposition S3 (Alignment and projected-residual bound). *Let $F(\rho) = \sum_i \pi_i \ell_i J(\rho_i)$ subject to $\sum_i \pi_i \rho_i = B$ and box bounds. If $-\ell_i J'' \geq m > 0$, a feasible $\tilde{\rho}$ is optimal iff there are β and $s_i \in N_{[\underline{\rho}_i, \bar{\rho}_i]}(\tilde{\rho}_i)$ with $\ell_i J'(\tilde{\rho}_i) - \beta - s_i = 0$. For any such closest normal-cone elements, $r_i = \ell_i J'(\tilde{\rho}_i) - \beta - s_i$ satisfies*

$$0 \leq F(\rho^*) - F(\tilde{\rho}) \leq \frac{1}{2m} \sum_i \pi_i r_i^2.$$

Proof. Strict concavity makes the feasible optimum unique and KKT conditions necessary and sufficient. With $\Delta_i = \rho_i^* - \tilde{\rho}_i$, strong concavity gives $f_i(\rho_i^*) - f_i(\tilde{\rho}_i) \leq f'_i(\tilde{\rho}_i) \Delta_i - m \Delta_i^2/2$. The budget equality removes the β term, normal-cone monotonicity gives $s_i \Delta_i \leq 0$, and $r_i \Delta_i - m \Delta_i^2/2 \leq r_i^2/(2m)$. Summing proves the bound. $\square$